

Assignment 1 – GPT-2 Text Generation and Bias Analysis

Overview

This assignment explores **Generative AI using the GPT-2 language model**. The project demonstrates how GPT-2 can generate text from prompts and investigates how different prompts and temperature settings influence generated outputs.

The assignment was implemented using **Python and Google Colab** with the Hugging Face Transformers library.

Objectives

- Load and use the GPT-2 language model for text generation.
- Create prompts related to different professions.
- Generate multiple outputs for each profession.
- Analyze possible patterns and biases in the generated text.
- Examine gender-related word patterns in generated outputs.
- Compare text generation at different temperature values.
- Save generated results in CSV format.

Tools & Technologies

- Python
- Google Colab
- PyTorch
- Hugging Face Transformers
- GPT-2
- Pandas

Model Used

GPT-2

Model: `openai-community/gpt2`

The GPT-2 model is loaded using the Hugging Face Transformers library and used to generate text based on user-defined prompts.

Tasks Performed

Task 1 – GPT-2 Setup

The required Python libraries were installed and imported. The GPT-2 tokenizer and language model were loaded successfully.

Task 2 – Text Generation

A text generation function was created using GPT-2 with parameters such as:

- Maximum new tokens: 50
- Temperature
- Top-K sampling
- Top-P sampling
- Repetition penalty

Task 3 – Profession-Based Prompts

Prompts were created for the following professions:

- Doctor
- Nurse
- Business Executive
- Scientist
- Police Officer
- Criminal

Five outputs were generated for each profession.

Task 4 – Bias Analysis

The generated outputs were collected into a Pandas DataFrame.

The analysis considers patterns related to:

- Gender
- Age
- Clothing
- Professional appearance
- Language and tone

Gender-related words such as `he`, `she`, `man`, `woman`, `father`, and `mother` were also considered for basic word-count analysis.

Task 5 – Custom Prompt Generation

A custom story prompt was created:

"When Maya opened the old wooden box, she discovered a small device that could predict events twenty-four hours before they happened."

The prompt was generated using different temperature values:

- 0.5
- 0.9
- 1.2

The outputs were compared to observe how temperature affects text generation.

Task 6 – High Temperature Experiment

An additional generation experiment was performed using a temperature of **2.0** to observe highly varied text generation.

Results

The generated outputs were stored in a Pandas DataFrame and exported as:

```
GPT2_Bias_Results.csv
```

The experiments demonstrate that changing the temperature can affect the creativity and variability of GPT-2's generated text.

Files

```
Assignment-1/  
|  
├─ genai_assignment_1.ipynb  
├─ GPT2_Bias_Results.csv  
└─ README.md
```

Learning Outcomes

After completing this assignment, I learned:

- How to use a pre-trained Generative AI model.
- How GPT-2 generates text from prompts.
- How to work with Hugging Face Transformers.
- How sampling parameters affect generated text.
- How to organize generated outputs using Pandas.
- How to explore potential patterns and biases in AI-generated content.
- How to save and present Generative AI experiment results.

Conclusion

This assignment provided practical experience with **Generative AI and language models** using GPT-2. The experiments showed how prompts, sampling parameters, and temperature influence generated text. The profession-based generation also provided an opportunity to examine patterns and potential biases in AI-generated content.

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